Muhammed Emin ilhan
150719513

Linear Algebra
—FINAL EXAM—

"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues."



QUESTION 1:

PART A) $A=$ ① $\begin{bmatrix} 1 & \alpha & -4 & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \rightarrow A=$ ② $\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 2 & 28 & -12 & 4 \end{bmatrix}$

$A=$ ③ $\begin{bmatrix} 1 & \alpha & -2\gamma & 2\alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 0 & 28-\beta & -12 & 4 \end{bmatrix} \rightarrow A=$ ④ $\begin{bmatrix} 2 & 2\alpha & -2\gamma & 2\alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 0 & 28-\beta & -12 & 4 \end{bmatrix}$

$A=$ ⑤ $\begin{bmatrix} 0 & 2\alpha-\beta & -2\gamma-3 & 2\alpha-\alpha^2 \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 0 & 28-\beta & -12 & 4 \end{bmatrix} \rightarrow A=$ ⑥ $\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 0 & 28-\beta & -15 & 4-\alpha^2 \end{bmatrix}$

Muhammed Emin Ilhan
150719513

Linear Algebra
FINAL EXAM

1. sorunun devamı: $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & \alpha & 0 & 1 \end{bmatrix}$

A PARTI

$K = 28 - \beta$

$\beta = -1$

$\alpha = 0$

$-4 = 0$

$\alpha = 0$

$K = 28 - (-1) = \underline{\underline{29}}$

$\alpha = 0$ $\quad m = 0$

$-4 = 0$

$K = 29$

$\beta = -1$

QUESTION 1:

PART B)

$U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 2 & -1 & 3 & 0 \\ 1 & 14 & 6 & 2 \end{bmatrix}$ ① $\longrightarrow$ $U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 2 & -1 & 3 & 0 \\ 1 & 14 & -6 & 2 \end{bmatrix}$ ②

$U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 10 & 0 & 2 \\ 2 & -1 & 3 & 0 \\ 1 & 14 & -6 & 2 \end{bmatrix}$ ③ $\longrightarrow$ $U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0^{-1} & 10^{-14} & 0-6 & 2-2 \\ 2 & -1 & 3 & 0 \\ 1 & 14 & -6 & 2 \end{bmatrix}$ ④

$\longrightarrow$ $U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & -4 & -6 & 0 \\ 2 & -1 & 3 & 0 \\ 1 & 19 & -6 & 2 \end{bmatrix}$ ⑤

1. SORUNUN DEVAMI:

PART B):

$$U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & -4 & -6 & 0 \\ 2 & -1 & 3 & 0 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & -18 & 0 & 0 \\ 2 & -1 & 3 & 0 \\ 1 & 19 & -6 & 2 \end{bmatrix}$$

= upper triangle matrix

PART C):

$$= \begin{bmatrix} 1 & \alpha & -4 & \alpha \\ 3\alpha & 5 & -94 & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 2 & -1 & 3 & 0 \\ 1 & 14 & -6 & 2 \end{bmatrix} = \rightarrow \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 2 & -1 & 3 & 0 \\ 0 & 29 & -15 & 4 \end{bmatrix}$$

Muhammed Emin Ilhan
150719513

Linear Algebra
Final Exam

## QUESTION 2:

PART A): given $x = 5y$

$\Rightarrow x - 5y = 0$

- This is four dimensional geometry, These can be written 2 different variable: a and B

$$K = \left\{ \begin{bmatrix} 5 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

PART B): $Rank(A) = 4$

$Ax = B$, unknown geometry to find solutions

$$K = \begin{bmatrix} 5 & 5 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

PART C): $x + 4y - z = 6$, $m = 1$, $n = 9$ and $\Phi = -1$

$Ax = B$ don't have any solution

$x \in R^5$ for $Ax = B$, The left hall space

- A hal dimension inside:

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 0 \\ -1 \\ 0 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \\ 0 \\ 6 \end{bmatrix} \right\}$$

There are 5 dimensions space here.

Muhammed Emin ilhan
150719513

Linear Algebra
FINAL EXAM

## QUESTION 3 :

PART A) : $H^2 = H \cdot H = (I - 2UU^T)(I - 2UU^T)$

$= I^2 - I \cdot 2UU^T \cdot I + 4UU^T \cdot UU^T$

$= I(I - 2UU^T - 2UU^T) + 4U \cdot \|U\|^2 \cdot U^T$

$= I - 2UU^T - 2UU^T + 4U \cdot \|U\|^2 \cdot U^T$

$\|U\|^2$ is equal to $= U^T \cdot U$

$\|U\|$ is equal to $= \pm\sqrt{U^T U}$

$\|U\| = 1$

That's why, $H^2 = I - 4UU^T + 4UU^T$

$H^2 = I - 4(1) + 4(1)$

$\rightarrow H^2 = I$ //

PART B) : $HH^T = (I - 2UU^T)(I - 2UU^T)^T$

$= (I - 2UU^T) \cdot (I^T - (2UU^T)^T)$

$= (I - 2UU^T) \cdot (I - 2(U^T)^T U^T$

$= (I - 2UU^T) \cdot (I - 2UU^T)$

$= I^2 - 2UU^T - 2UU^T + 4UU^T \cdot UU^T$

$= I - 4UU^T + 4U\|U\|^2 U^T$

$= I - 4UU^T + 4UU^T$

$= I - 4(1) + 4(1)$

$HH^T = I$ //, so, $H^T H = I$

$HH^T = H^T \cdot H = I$

so H is an orthogonal matrix. //

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Muhammed Emin ilhan
150719513

Linear Ajebra
FINAL EXAM

QUESTION 3 :

PART C) : For a matrix A, a vector (whic is non zero) will be eigen vector (suppose, v) of A corresponding eigen value A if, $\boxed{AV = \lambda V}$

so its $= HU = (I - 2UU^T).U$

$= IU - 2UU^T.U$

$= U - 2U.\|U\|^2$ As I is identity matrix.

- I değerinin sadece bir tane eigen değeri vardır ;

$IU = I.U = U$ (also), $\|U\| = \pm\sqrt{U^T U}$

- where, U is an unit vector, so $\|U\| = 1 = \pm\sqrt{U^T U}$

Finally, $U^T.U = \|U\|^2 = 1$

$\rightarrow HU = U - 2U.1 = -U$

$HU = -U = (-1).U$, so we have : $HU, \lambda U$

where : $\lambda = -1$

That's why, U is an eigen vector of H and corresponding eigen value for U is $= \underline{\lambda = -1}$

Muhammed Emin ilhan
150719513

Linear Algebra
FINAL EXAM

## QUESTION 4 :

PART A) : $v(0) = 10$
$w(0) = 40$ is given;

$$\begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix}$$

$$= \begin{cases} V_{k+1} = -2V_k + W_k \\ W_{k+1} = V_k - 2W_k \end{cases}$$ differential equation getting expass the process.

PART B) : 1 saniye sonra, 1. tank :

$$V_{1+1} = -2V_k + W_k$$

$$V_2 = -2 \cdot 10 + 1$$

$$= -20 + 1 = 19L$$

Yüzdelik hesabı için; $\frac{19}{10} \cdot 10\cancel{0} = =$ %190

PART C) : 25 Litter remaines at steady stade.

-